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SuRE WP6: PU UV (preliminary) results

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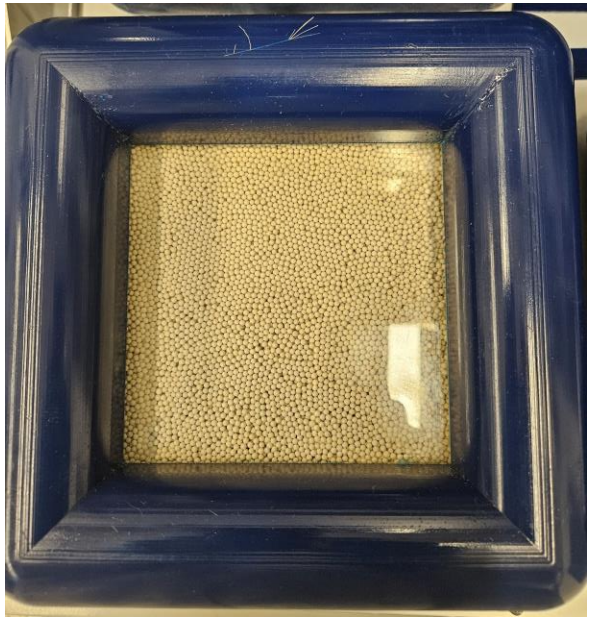
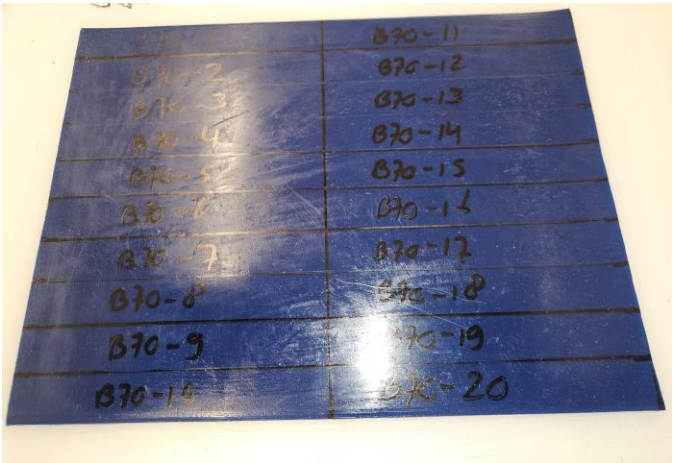
Sample overview

- 12x minipatches, 3 different hardnesses
  - 4x 90A (Serial# 107 – 110)
  - 4x 80A (Serial# 206 – 209)
  - 4x 70A (Serial# 307 – 310)
- 4x 20x15x0.2 cm^3 PU sheets
  - 2x B 70A
  - 1x B 80-85A
  - 1x C+ 85A (lower quality product)
  - ➢ Three unique sheets cut in 20 10x1.5 cm^2 strips for tensile testing

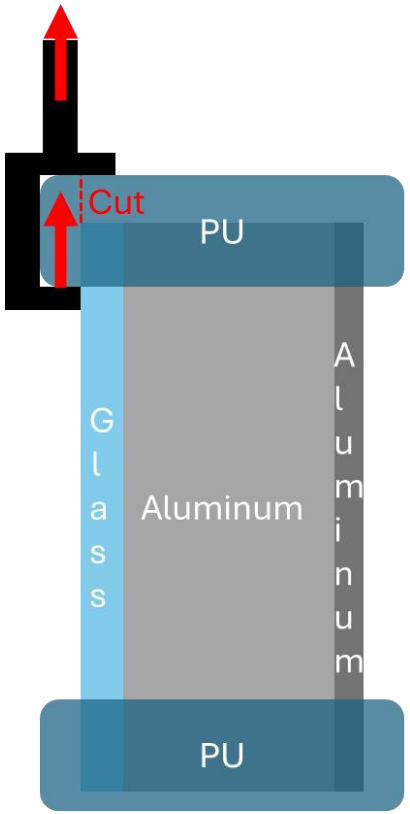


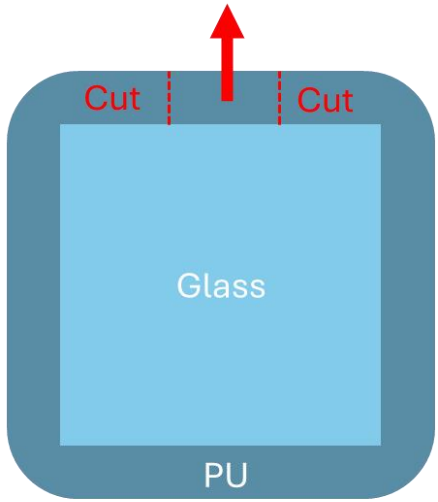


Testing plan

- Goals:
  - Test PU tensile strength and PU-glass and PU-Al adhesive strength as a function of UV exposure
  - Measure moisture ingress in minipatches after humidity exposure
- Strength testing: tensile for PU, lap shear strength for minipatches
  - Sample cutting of minipatches to be done after UV exposure
- UV exposure: 1000h at condition A3 (see table below)
  - Standard test for PV components, previously performed on dogbones and edge sealant samples in MariSol, but now with RH = 80% instead of 20% to test moisture ingress
  - Black panel T means that black item in chamber reaches 90 C. PU will be close to that since dark blue
    - Many other PV tests with max ambient T of 85 C, so components should be stable at such T
    - 1000h UV dose (kWh/m^2) about 2 years of exposure in Sætre, but test at higher T (accelerates degradation)

Condition #	T_chamber[C]	Black panel temp [C]	Irradiance @340nm [W/m2 nm]	RH [%]
A1	45	70	0,8	20
A2	55	80	0,8	20
A3	65	90	0,8	20
A4	75	100	0,8	20
A5	85	110	0,8	20





UV chamber test

- Samples in UV (top image is before start):
  - 5 minipatches: 2x 90A, 1x 80A, 2x 70A
  - 9 tensile samples per sheet type (27 total)

- Test paused after 209.09h of exposure for inspection (bottom image):
  - ➢ All samples (both minipatches and tensile strips) severely darkened
  - ➢ Several of the minipatches displaying burn marks

➢ <u>UV test stopped completely as result of the observations</u>

UV results after 209h: minipatches

- All samples darkened, and several with burn marks
- Clear differences in visual appearance of PU types (inspection some hours after test stop, all samples at RT):
  - 90A (107&108): top black but no burn marks. PU still feels hard to touch. Some sides still blue
  - 80A (206): top black and burn marks. PU on top deforms on touch (feels soft). Sides also black.
  - 70A (308 & 309): top black and most severe burn marks. PU deforms easily on touch (feels softer than 80A). Sides black as well.
- Due to sample height, samples closer to light source, leading to elevated irradiance and temperature
  - Technical support: irradiance on the minipatches is about 25% higher than on the tensile ones
    - Irradiance 1 W/m^2
  - Based on irradiance increase, approximate new T is ~95 C for black samples [Stefan-Boltzmann energy balance approximation]
    - Measure manually to confirm
    - ➢ Likely higher than what will be experienced in the field

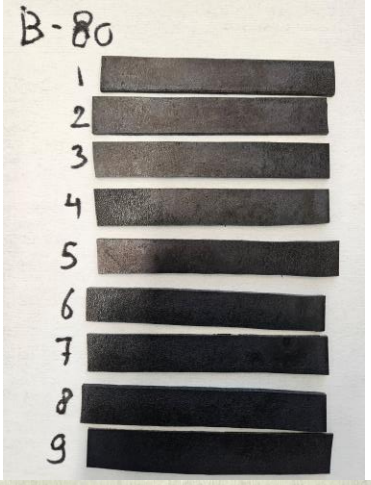
UV results after 209h: minipatches (2)

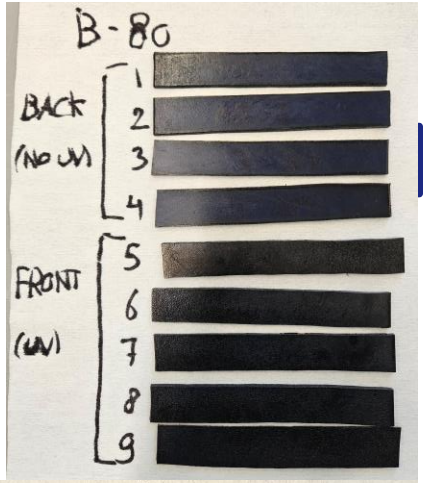
- Gravimetric measurements:
  - Ref samples no mass increase
  - UV samples increase between 2 and 8g
    - Possibly lower ingress with decreasing sample hardness, but differences too small to say
  - ➢ Possibility of moisture ingress, but ingress too low for conclusive results (longer measurement series needed)

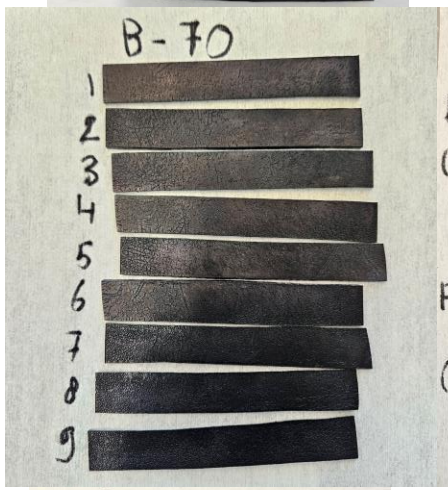
Sample #	Mass pre-UV	Mass post-UV (g)	Difference (g)
107	3312	3318	6
108	3316	3324	8
109 (REF)	3320	3320	0
206	3258	3264	6
308	3252	3256	4
309	3242	3244	2
310 (REF)	3268	3268	0

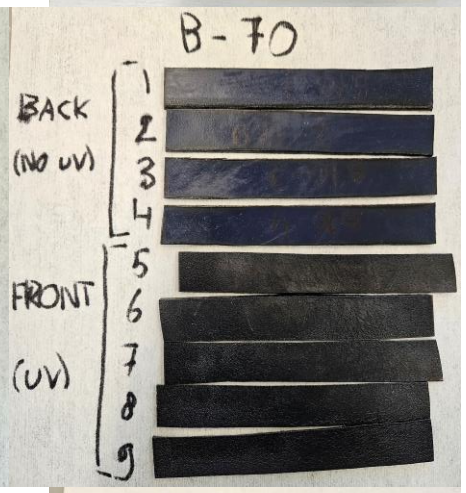
UV results after 209h: tensile strips

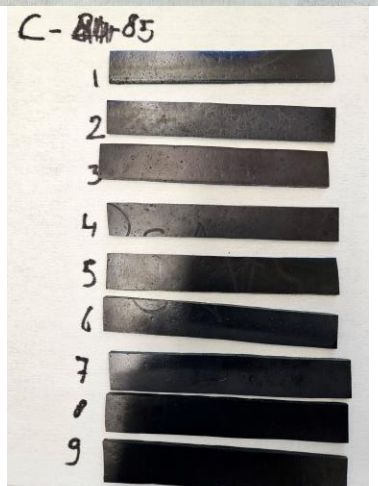
- Samples at same level as trey, so irradiance and sample T not higher than test standard (T up to 90 C).
- Darkening and cracking of samples observed, but no burn marks
- Also clear visual differences between sample types:
  - Images on left are front of all samples after UV, images on right are with first 4 samples flipped to show back (original color)
  - B 80-85A: samples black with small cracks
  - B 70A: samples black with (seemingly) larger cracks
  - C 85+: samples black but no cracks
  - ➢ C 85+ supposedly lowest quality but visually look best
- Tensile testing not done yet

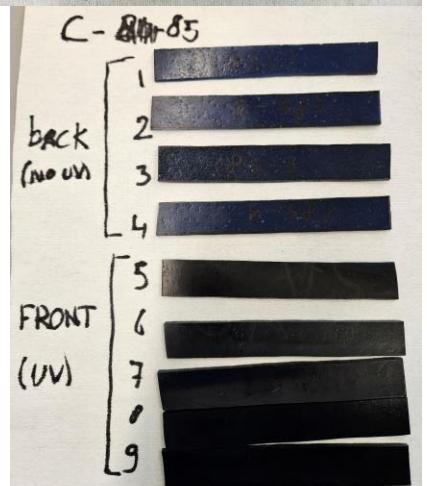












Suggested way forward

- New UV test at lower T to avoid burning
  - Suggest condition A1 (chamber T 45, black panel T 70 C)
    - Still somewhat elevated temperature to accelerate UV degradation, test at RT would limit degradation severely
    - Max temperature on minipatches expected to be ~75 C
      - NB! Will be measured before test is started
  - Samples in UV:
    - Minipatches: use 'fresh' ones. Not much use for exposed ones in new UV test.
      - Total matrix:
        - Reference (no UV): 1x 90A, 2x 80A, 1x 70A
        - 209h UV at A3 condition: 2x 90A, 1x 80A, 1x 70A
        - 1000h UV at A1 condition: 1x 90A, 1x 80A, 1x 70A
    - Tensile samples: Take out 6 samples per PU type to test now, leave 3 in per type. Put in 5 fresh ones per type
      - Total matrix (per PU type):
        - Reference (no UV): 6 samples
        - 209h UV at A3: 6 samples
        - 1000h UV at A1: 5 samples
        - 209h UV at A3 + 1000h at A1: 3 samples
      - ➢ Have second B70 sheet to include more samples as reference/1000h UV at A1, and to assess effect of pull testing

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Relevant information from manufacturer

- To avoid many (expensive) iterations of testing with more surprises, it would be good to get as much information on the PU as possible from the manufacturer.

- Questions to ask:
  - What is the expected thermal stability of the PU? At what upper T limit can stress testing be done?
  - Is there information (theoretical and/or testing results) regarding UV stability of the different PU types?
  - Under what sample and test conditions (e.g. sample dimensions, pull rate, ambient T, potential preconditioning) was the strength of the 90A PU (image on right) evaluated?
  - Are there any other material parameters that can be shared on the material?
  - What was the motivation to pick this material for the hinges compared to other alternatives?

Property	Value
Linear Elastic	
Angle Of Friction	
Bulk Modulus	
Compressive Strength	67,60 MPa
Density	1100,00 kg/m^3
Fracture Toughness	
Poisson Ratio	0,49
Shear Modulus	
Stiffness	
Ultimate Strain	
Ultimate Tensile Strength	75,45 MPa
Yield Strength	100,00 kPa
Young's Modulus	41,33 MPa